

Steven Samu

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Education

McMaster University, B.Eng in Computer Engineering

Sept 2020 – May 2025

- Cumulative GPA: 3.7/4.0

Technical Skills

Languages: Python, C & C++, Shell, System Verilog & Verilog, Perl, Makefile, Java

Software's and Applications: Linux & Unix OS, SoC, VLSI, Git, Perforce, Matlab, RTL Design, HPC, $\LaTeX$, Docker

Work Experience

SoC Integration & Verification Engineer Internship, AMD – Markham, ON

May 2023 – May 2024

- Performed RTL integration of System Management Unit (SMU) & Control Fabric (CF) IPs for AMD's **MI Instinct** chiplet-based accelerators. Developed automated **Formal Verification** and **Logic Equivalence Checking** (LEC) frameworks using **Python** and **Shell** scripting, accelerating verification workflows by **>20 engineering hours** per IP unit.
- Developed and maintained clock unification frameworks to harmonize multiple clock domains across chiplet boundaries. Lead integration of the new Mid System Management Unit (MSMU) microsequencer, introduced to optimize latency for SMN control transactions. Implemented comprehensive signal integrity verification using **System Verilog** and **UVM** methodologies in AMD's custom **Linux** environment.
- Ensured end-to-end integration correctness of accelerator IP blocks from initial RTL import through near tape-out readiness. Developed an automated regression testing infrastructure using **Python** and **C++**, achieving significant reduction in integration validation time between project milestones.

EDB Mainframe Developer Internship, TD Bank – Toronto, ON

May 2022 – Sept 2022

- Worked in a pod based team to develop **SAS** & **COBOL** tools to extract data from national account data sets for further analysis and problem detection. Under tight deadlines, mass recalled archived data sets and verified them for internal investigations.
- Reformatted EDB team-wide confluence to include detailed pages with critical information on critical engineering tools, API server info and necessary **Agile** documentation. Programmed data parsing tool to track number of monthly completed pod tasks from the teams **Jira** Databases utilizing **JQL**.

Academic Projects & Research Experience

YOLO Object Detection on Raspberry Pi - Python, Yolo 11, Docker, Linux

Supervisor: Dr. M.Hassan

- Implemented and optimized YOLO 11 object detection **Python** model for embedded drone applications using the VisDrone dataset, achieving **85% confidence** threshold for accurate object classification. Developed performance optimization techniques through NCNN sub-formatting, doubling computation speed while maintaining prediction accuracy on **Raspberry Pi 4** hardware.
- Engineered complete video processing pipeline for real-time object detection, implementing H.264 encoding for optimal quality-performance balance. Developed custom Docker containerization solution to interface **Pi Camera** with YOLO detection framework in resource-constrained environments.

OoO Execution Modeling Cache Simulation - C++, C, Python

Course: Comp Eng 4DM4

- Implemented Out-of-Order (OoO) Execution in McMaster's **Octopus** cache simulator by designing and integrating a Re-Order Buffer (ROB) and Load-Store Queue (LSQ). Enhanced the given **C++** codebase to optimize instruction scheduling while maintaining precise in-order retirement semantics.
- Achieved **>25% improvement** in simulation performance through advanced **computer architecture** optimizations. Developed comprehensive logging infrastructure to enable cycle-accurate analysis of cache behavior and validate architectural correctness. Leveraged **clangd** for streamlined code navigation and formatting, improving development efficiency and understanding of simulator transactions.

Hardware Implementation of Image Decompressor - Verilog

Course: Comp Eng 3DQ5

- Designed a digital system focused on image compression of .mic16 files. Compressed data for a 320 x 240 pixel image to be delivered via UART interface to external SRAM within the courses **FPGA** hardware. Leveraging skills in **Verilog** and **RTL** design, designed a system that achieved near lossless decoding, signal transformation, interpolation and color space conversion.
- Design achieves the projects constraint of MUX hardware efficiency and the constraint of limited space. By computing the up sampling and color space conversions calculation using only three MUX's, processed each individual pixel's RGB value to be sent to the SRAM.

Software Defined Radio (SDR) - Python, C++

Course: Comp Eng 3DY4

- Developed a SDR application to receive real-time FM mono/stereo audio and digital data using the radio data system (RDS) protocol from a front-end radio frequency (RF) hardware. Project developed and ran from a Raspberry Pi coupled with a RF dongle.
- SDR system prototyped, built and tested in **Python**. Later converted and re-factorized into **C++** to leverage the **C++** vector and thread libraries to achieve real-time FM processing. Real time radio signals were received, processed, and listened to at end of project from local stations.

Personal Projects

Custom Built PC

- Architect and built custom high-performance PC, implementing strategic hardware selection and bi-annual upgrades tailored for gaming. Optimized system around AMD Ryzen platform and NVIDIA's RTX architecture to leverage their advanced rendering techniques including real-time light simulation and dynamic shader optimization.

Additional Experience's and Awards

Awards: McMaster University Dean's Honor List 2020/21/22 & President's Entrance Award

McMaster Google Developer Student Club

- Community & Code Team Member (2021)

Google Solutions Incubator Program

- Business Analyst & Designer (2021)

Hobbies and Interests

Rock Climbing

- Fully certified indoor sport climber, advid boulderer and aspiring big wall climber. Climbing is the most important and consistent hobby in my life, allowing me to simultaneously train both my brain and my body.

Video Games

- Gaming has been a consistent hobby of mine, starting on handhelds as a child to a now advid PC gamer. Construction & Management games (Factorio, Satisfactory) are favorites of mine.

Relevant Courses

Math 3QC3 - Introduction to Quantum Computing

- Gained foundational knowledge of quantum computing, including the mathematical framework, qubits, and quantum gates. Analyzed and synthesized quantum circuits, with focus on Shor's algorithm to demonstrate key quantum principles.
- Studied quantum complexity theory and parallel computing principles through hands-on algorithm implementation. Demonstrated proficiency in quantum information theory by analyzing quantum parallelism and its applications in computational speedup.

Applications of Linear Algebra
Numerical Methods for Scientific Computing
Advanced Internet Communications
Advanced Probabilities & Random Processes

Computer Communication Networks
Computer Architecture
Control Systems Design
Algorithm Design and Analysis

Digital Systems Design
Logic Design
Operating Systems

Circuits and Waves
Embedded Systems
Communication Systems